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Scaffolding Children's critical reflection on intelligent technologies: Opportunities from speculative fiction

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A B S T R A C T

Current technological development of Artificial Intelligence (AI) requires educational practices that address the social and ethical implications derived from these emerging technologies. To this end, an increasing number of educational practices are pursuing the goal of supporting children's critical reflection on these topics. Our research aims at understanding how speculative fiction-based resources can meet and respond to the goals of supporting children's critical reflection on AI technologies and their impact on society. Through revisiting relevant literature on these topics and critically analyzing our own practices in three different settings, we identify a set of opportunities and challenges oriented at guiding the design of resources capable of taking advantage of speculative fiction as a way to support critical reflection.

1. Introduction

As robots, automated systems, and AI evolve and gradually enter everyday life, increasingly integrate into everyday life, there is growing interest in enhancing children's literacy on these intelligent technolo-gies. As a starting point of this research, several researchers are building frameworks and guidelines to define the scope of AI literacy. Research in this field is well exemplified in Touretzky et al. (2019), who identify five big ideas to guide the design of K-12 curriculums. This contribution is further expanded by Long and Magerko (Long & Magerko, 2020) who outline sixteen key competencies that are considered necessary for AI literacy (e.g. *being capable of recognizing AI, imagining possible future ap-plications of AI, understanding what knowledge representation is, identifying and describing different perspectives on the key ethical issues surrounding AI,* etc.).

The definitions of AI literacy proposed by Touretzky et al. (2019) and Long and Magerko (2020) highlight the complex and transdisciplinary nature of this field, which encompasses both supporting children in understanding the technical functioning of AI (Brummelen, Heng, & Tabunshchyk, 2021; Druga et al., 2019, pp. 104–111; Hitron et al., 2019; Lin et al., 2020) and fostering their ability to critically reflect on AI's societal impact (Bilstrup et al., 2020, pp. 1211–1222; Coenraad, 2022; Payne, 2020; Schaper et al., 2020, pp. 1–4).

This latter aspect echoes concerns raised by several researchers regarding the limitations of mainstream approaches to digital literacies. Specifically, critics argue that digital literacy practices are often framed through overly technical and uncritical perspectives, neglecting inquiry, analysis, critique, and socially engaged viewpoints (Fuchs, 2021, pp. 1–440; Hinrichsen & Coombs, 2014; Pangrazio, 2016). In response to this gap, an increasing number of educational projects are starting to move computer science education beyond computational thinking-focused approaches to incorporate key contents and compe-tencies related to supporting children's critical reflection on intelligent technologies (Bilstrup et al., 2020, pp. 1211–1222; Coenraad, 2022; Payne, 2020; Schaper et al., 2020, pp. 1–4). However, as Iivari et al. (2022) highlight, the application of critical methods remains insuffi-ciently explored in Child-Computer Iinteraction research. In this sense, the authors call for more research that draws from speculative design, design fiction, and reflective design approaches which can allow chil-dren to critically examine the status quo, challenge cultural norms, and propose alternative values and ideas of emerging technologies. Building on this call, our article addresses a research field that we broadly

grouped under the umbrella term of *Reflective AI Literacy (RAIL)*. Specifically, we consider RAIL practices as educational activities that explicitly aim to support students' capacity to critically reflect on the design, development, and use of intelligent technologies, taking into account their embeddedness in social structures, discourses, and practices.

The emerging nature of this field presents a wide array of challenges. First, addressing competencies related to critical reflection can become complex and ambiguous, since multiple meanings have been attributed to this concept, transforming it into a sort of buzzword used ubiquitously (Edelsky & Cherland, 2006). Second, the fast pace of innovations in AI complicates efforts to clearly define the topics that need to be addressed and the competencies that should be supported. Third, the emerging nature of this field necessitates further exploration of methods, tools, and strategies that could be particularly suitable to this end. These challenges inevitably open questions for researchers and educators interested in developing and designing resources for RAIL, as evidenced by the increasing number of activities aimed at supporting sharing resources in this field (Charisi et al., 2020, pp. 89–95; Sharma et al., 2022, pp. 1–6).

To support researchers and practitioners in designing RAIL activities, this article pursues three main objectives. First, we aim to identify different (and often complementary) dimensions through which critical reflection on AI can be framed. We achieve this by unpacking different epistemological and pedagogical traditions related to the notion of critical reflection and mapping them with different RAIL's initiatives for children. Second, we aim to deepen the reflection on appropriate methods and tools to support RAIL. Specifically, our research investigates the potential of Speculative Fiction (SF) techniques to facilitate critical reflection on intelligent technologies and their impact on society. This is examined through a reflective analysis of three projects in which we employed diverse SF-based techniques to support children's reflection on intelligent technologies. Third, by integrating the revision of different approaches to critical reflection, insights from our practices, and examples from SF narratives, we identify a set of opportunities and challenges related to using SF in RAIL projects.

In this regard, this article provides a threefold contribution. First, it offers valuable perspectives for researchers and practitioners broadly interested in RAIL, providing analytical lenses to examine and design RAIL activities for children. Second, it highlights the opportunities and challenges of using SF in RAIL practices, offering a foundational reference for those particularly interested in exploring SF's potential as an educational tool. Third, this article demonstrates how combining theoretical analysis, empirical reflection, and insights from fictional narratives can serve as an effective methodological approach in educational practices.

2. What does it mean to support children's critical reflection on AI?

The first step in discussing children's critical reflection on AI is to examine the conceptual, theoretical, and epistemological foundations upon which these practices are or can be built. In other words, we need to ask ourselves which discourses shape our practices and how we define the notions of *critical reflection* and of *critical reflection on AI.* To address this, we first need to unpack the concept of critical reflection.

The historical roots of critical reflection trace back to Socratic teaching, which 2500 years ago, employed argumentative dialogues to stimulate critical thinking and uncover underlying presuppositions. This long standing trajectory shapes critical reflection as a key competence in human reasoning as well as a crucial quest in pedagogy. Although a deep historical analysis of this concept falls beyond the scope of this article, we briefly outline key theoretical and pedagogical contributions to distill guiding principles for framing RAIL practices.

Johnson and Morris (2010), in their framework for critical citizenship education, identify two primary traditions related to critical reflection. The first includes pedagogical practices that build on a notion of *critical thinking* akin to the ideas proposed by Dewey (1997) and by Mezirow (1987). The second is rooted in frameworks derived from Critical Theory, Frankfurt's school and critical pedagogies approaches (Freire, 1970; Giroux, 2020). Widening this perspective, a third complementary approach emerges from the works of contemporary feminists scholars (Braidotti, 2013; Haraway, 2016) and art and design practices (Kozubaev et al., 2020, pp. 1–15) which emphasizes the creative dimension of reflection and its quest for diverging or disrupting normative ways of doing.

The first approach conceptualizes critical reflection primarily as an individual cognitive process. Following Dewey's definition, reflective thinking is considered as an *"active, persistent and careful consideration of any belief or supposed form of knowledge in the light of the grounds that* support *it, and the further conclusion to which it tends"* (1997). Similarly, Mezirow (1987, pp. 214–216), emphasizes the self-corrective nature of this process, defining critical reflection as the examination of the presuppositions underlying one's beliefs. Accordingly, critical reflection involves becoming critically aware of our own assumptions, questioning the premises on which problems are defined and challenging the validity of presuppositions in prior learning. This implies competences such as analyzing one's own positionality, recognizing inherent biases (de Andreotti, 2014) and inhabiting multiple, potentially contradictory perspectives and worldviews related to a topic (McLaughlin, 1992). In this view, critical reflection functions as a self-corrective process that activates a broad range of meta-cognitive skills, including the ability to embrace doubt, unlearn ingrained ideas, and develop an inquiry-driven mindset.

The second approach (Freire, 1970; Giroux, 2020) takes a distinct perspective by emphasizing the socio-political dimensions of critical reflection and its role in social transformation. Like the previous approach, it underscores the need to not take things for granted, questioning assumptions and looking beneath the surface. However, it further asserts that students must develop a critical knowledge of society, its political and economic dimensions, its power and authority dynamics, and its role in perpetuating privileges and inequalities. This perspective encourages the critical analysis of social, political, and economic structures to see beyond the surface (Westheimer & Kahne, 2004) and the need for understanding the interconnections between values, culture, power and social justice (Giroux, 1980). Furthermore, it advocates for incorporating marginalized perspective to challenge the dominant discourse (Parker, 1996) and fostering active civic engagement (Giroux, 1980). From a pedagogical standpoint, this approach promotes competencies such as critically assessing societal issues, investigating historical trajectories that shape current conditions, and identifying and challenging power imbalances. Furthermore, it aims at enhancing competences related to people's transformative agency (Iivari, Iversen, et al., 2024, pp. 322–337) and their capability to advocate for the implementation and enhancement of current practices in various contexts.

Finally, a third approach, deeply interconnected with the previous two, focuses on fostering divergent and imaginative thinking. Building on the previous approaches, it supports a process in which children first become aware of their assumptions and social injustices and then creatively explore ways to transform them. In other words, critical approaches to reflection involve both deconstructive and constructive efforts: deconstructive efforts challenge contradictions and injustices, while constructive efforts encourage imagining and developing alternatives (Murray & Ozanne, 2006). These relations are well described in Lipman's notion of complex thinking (Lipman, 1976) which encompasses three different dimensions: critical thinking, caring thinking, and creative thinking.The creative dimension involves imagining, inventing, breaking schemas, and envisioning possibilities beyond established norms. Contemporary feminist and posthumanist scholars further develop this perspective by emphasizing the need for disrupting normative ways of thinking and doing. For instance, Donna Haraway

(2016), proposes replacing the notion of "reflection" with the concept of "diffraction" to address the need for undoing our usual ways of thinking and finding new stories and images to think-with. Similarly, Rosi Braidotti (2013) advocates for the loss of familiar habits of thoughts "*in order to pave the way for creative alternatives*" (pp.88-89). These perspectives introduce a crucial dimension to critical reflection—one that not only critiques existing paradigms but also actively envisions and constructs radically new possibilities. From a pedagogical standpoint, this third approach fosters competencies such as: challenging dominant paradigms and norms, thinking beyond present constraints, disrupting conventional perspectives, and imagining alternative futures.

This overview of existing approaches offers lenses through which critical reflection on AI can be framed, specifically.

(1) *Critical thinking:* The ability to self-examine, challenge, and potentially reconsider one's own assumptions and beliefs.
(2) *Critical understanding*: The capacity to analyze socio-political systems, recognize their implications and their role in perpetuating privileges and oppressions, and engage in efforts toward social transformation.
(3) *Critical Imagining:* The ability to challenge existing paradigms and envision unestablished and unconventional possibilities.

In the following section, we will explore how these dimensions have been integrated into educational practices designed to support children's reflection on AI.

2.1. Critical reflection on AI and critical thinking

Initiatives that address children's critical reflections on AI through the lens of *critical thinking* generally pursue the following aims.

(1) Fostering children's capacity to question their pre-established assumptions, premises, and discourse embedded in intelligent technologies.
(2) Supporting children in understanding different viewpoints in relation to digital technologies.
(3) Encouraging children in being responsible and informed users as well as the need to empower them as critical and committed future designers and developers.

From an educational perspective, this notion is well described through the notion of *computational empowerment* (Dindler et al., 2020), where the authors emphasize the importance of guiding children through a design process that includes reflecting on the critical aspects of technology. This process helps children recognize the ethical dimensions of technology, the decisions involved in its design, and the potential consequences of those choices.

RAIL practices framed on the lens of *critical thinking* generally address two main goals. First, providing children with the technical knowledge and lenses to critically analyze the intelligent technologies that surround them and questioning what is taken for granted. A notable example is the "PB&J sandwich activity" from the AI ethics curriculum designed by Payne (2020). The authors design an activity based on children's everyday experience (i.e. asking them to define the perfect peanut butter sandwich) to prompt reflection on how algorithms define success. Similarly, another activity in the same curriculum, the "Cat and Dog" classification exercise, introduces children to issues related to biases in datasets.

Second, a complementary goal of these approaches is to foster children's self-reflection on their individual responsibilities as users and eventual future designers on intelligent technologies. Bilstrup et al. (2020) adopts an approach akin to the notion of *microethics,* as proposed by Herkert (2005), which examines ethics through the lens of individual responsibility in technological decision-making. To operationalize this, they develop a board game that prompts children to engage with critical questions about the ethical dilemmas and decision-making processes involved in designing machine learning-based applications.

2.2. Critical reflection on AI and critical understanding

Addressing critical reflection on AI through the lens of *critical understanding* entails an explicit focus on the socio-political dimensions of these systems and a a commitment to challenging and dismantling social and political inequalities derived from them. In this sense, the main goals of this approach are.

(1) Supporting the understanding of systems of power and how AI-based technologies may reinforce dominant interests (Luke, 2000).
(2) Recognizing the role of social and economic forces in shaping technologies and examining how technologies, in turn, shape society (Winner, 1980).
(3) Facilitating reflection on the role of technologies in perpetuating inequalities.
(4) Encouraging transformation oriented toward serving less powerful interests (Pandya & Avila, 2013).

This perspective is highly influential in science and technology studies (STS), with several authors analyzing the intertwined relationship between politics, economics, and technology development (Winner, 1980). In the field of Computer Science (CS) education, a key example of this approach is the work of Vakil (2018) who employs critical pedagogies and critical race theory to reframe CS education in ways that integrate ethics, justice, and equity both in curriculum design and learning experiences, as well as in policy-related decision-making. To this end, he introduces the notion of Justice-Centered Computing and stresses the need of providing students with knowledge on systems of power and politics in order to be capable of addressing ethics in CS education.

Within the field of RAIL practices, there are few examples that explicitly focus on critical understanding. One such example is Schaper et al. (2020) who designed activities to help children reflect on cultural diversity in AI. To achieve this, they invited children to organize a food celebration in a multicultural school, leveraging their knowledge of local food traditions to spark discussions about diversity and the geopolitical distribution of AI. Similarly, also Schaper et al. (2022) used value-drive design cards that displayed different technologies, actors, benefits and risks, and conflicts of interest as a way to enable students to map and identified possible value dilemmas related to intelligent technologies and reflect upon possible trade-offs that potential stakeholder may need to consider.

2.3. Critical reflection on AI and critical imagining

Practices that address Critical Imagining in relation to intelligent technologies aim at cultivating children's ability to break away from the established, embrace alternative perspectives, and imagine new possibilities in understanding, designing, and using these technologies. In the realm of art practice, design studies, and education, this approach addresses the quest for perspectives, methods, and tools that liberate us from our habitual patterns of thoughts and disrupt normative ways of doing, thereby expanding what is thinkable (Wilde et al., 2017). A compelling example of this perspective, can be found in the book "Ways of Being" (Bridle, 2022), which reframes AI as an invitation to explore non-human forms of intelligence and leverages this perspective to reformulate the ways we design intelligent technologies and how we relate to non-humans. On the other hand, in the field of design, examples can be found in research on "embodied design methods'" and "design futuring" approaches (Kozubaev et al., 2020, pp. 1–15) which makes use of either our bodies or futures explorations for defamiliarizing habitual ways of thinking and doing in order to pave the way for novel results.

Within the context of RAIL, providing children with alternative perspectives and ways of thinking has been explored through different approaches. For example, Iivari, Hartikainen, et al. (2024) employ various future-oriented design methods-such as design fiction, plus-and-minus scenarios, the miracle method, and design activism-to engage children in tackling bullying through the design of digital technology. Similarly, Ruppert et al. (2023) engaged a group of queer teenagers in AI learning through embodied tinkering, employing it as a strategic "knowledge-building praxis" within the context of Drag queen art and facial recognition technologies. Another noteworthy example is Bilstrup et al. (2022) who proposed the Embodied Data Modeling activity to support teenagers' understandingof personal data through bodily engagement.

2.4. The three dimensions as a starting point to reflect on practices and methods

The analysis of how critical reflection has been framed in AI literacy offers valuable insights for researchers and practitioners. First, it provides guidance for designing activities, tools, and resources by helping to clearly define the scope and goals of a RAIL project. For instance, a project aimed at supporting critical reflection on a topic such as "filter bubbles" (Parisier, 2012) and recommendation systems would be framed very differently according to the lenses employed. If it focuses on supporting *critical thinking,* it will mainly address goals related to promoting children's understanding of the working model of recommendation systems, making them aware and cautious about the biases in the information they may get and making them reflect on how to overcome these limitations in their usage of technology. Instead, if framed through the lens of *critical understanding*, it should also include a broader understanding of how filter bubbles may serve specific political and economic interests and how their functioning may perpetuate inequalities. Finally, if framed through the lens of *critical imagining*, it should help children to move beyond what is already there and think at these phenomena from diverging viewpoints (see Table 1).

Second, these lenses serve as valuable tools for examine existing practices, methods, and techniques to assess their adequacy to support specific RAIL goals. Building on this approach, in the following sections, we will use the identified dimensions as lenses to examine the potential of SF techniques to support children's reflection on intelligent technologies. We will begin by broadly contextualizing SF's potential to foster the different dimensions of critical reflection. Subsequently, we will analyze the potential of SF in our practices through the identified lenses.

Table 1
Summary of approaches to critical reflection.

Dimension	Main Aims
Critical thinking	- Fostering children's capacity to critically analyze the intelligent technologies that surround them and question their pre-established assumptions, premises, and discourse embedded in intelligent technologies - Supporting children in understanding different viewpoints in relation to digital technologies - Supporting children's self-reflection on their individual responsibilities as users and eventual future designers of intelligent technologies
Critical understanding	- Supporting the understanding of systems of power and how AI-based technologies may operate in the interest of power - Acknowledging the role of social and economical forces in shaping technologies and how technologies shape society - Facilitate reflection on the role of technologies in perpetuating inequalities - Fostering transformation-oriented toward serving less powerful interests
Critical imagining	- Breaking up with the established ways of thinking about and the design of intelligent technologies - Imagining other possibilities and other possible worlds, in terms of understanding, designing, and using these technologies

3. The potential of speculative fiction to support critical reflection

For the purposes of our research, we are particularly interested in exploring how SF practices can serve as valuable tools to promote children's critical reflection on intelligent technologies. From a broad perspective, SF aligns with emerging approaches, such as design futuring approaches (Howell et al., 2021) and fabulation based approach (Søndergaard et al., 2023, pp. 1693–1709) which seek to rethink design practices and design education.

The term "speculative fiction" was first coined by Robert Heinlein to distinguish the "more serious" science fiction narratives, i.e. those concerned with the impact of technologies on humans (Oziewicz, 2017). This notion was subsequently expanded thanks to the impulse of feminist science fiction of the 60s and 70s, which led to its redefinition.

Feminist science fiction writers positioned SF as a political space for challenging dominant structures and resisting patriarchal and colonial narratives. Moreover, they emphasized SF's potential as a "tool-to-think-with". For instance, Ursula LeGuin, through the notion of "Imaginative fiction" claimed the need for stories capable of training "*people to be aware that there are other ways to do things and other ways to be. That there is not just one civilization and it is good and it is the way we have to be*" (Curry, 2018). Donna Haraway (2016) further expanded this idea, introducing the notion of "*speculative fabulation*" as a way of thinking and narrating capable of creating new worlds through arousing an interest for what could be possible.

The evocative power of SF has made it a fertile ground for socially engaged practices that use fiction as a medium to open spaces for discussion and stimulating reflection of what is there and what could be possible. Examples of this approach can be found in literature, audiovisual production, art, design, and research, which are gradually transforming SF into a rich landscape of multimodal narratives. To cite a few, for instance, "The Camille Stories: Children of Compost" by Haraway (2016) employs speculative fabulation to imagine the symbiogenetic joining between humans and other species and questions our current relationship with the planet. Similarly, Le Guin (2015) in her book "The Dispossessed: an ambiguous utopia" explores the potential and limits of other socioeconomic and political systems. The tv-series "Black Mirror"' uses design fiction to problematize a near-future where technological innovations and human needs and instincts collide. In the field of design, Dunne and Raby (2013) shaped the fertile and evocative terrain of speculative design, giving rise to the creative and critical potential of speculative props. This potential was further expanded through the broad variety of "design futuring approaches" (Kozubaev et al., 2020, pp. 1–15), such as speculative design and design fiction (Bleecker, 2022) which combine speculation and design not only to solve immediate problems but to open spaces for reflection and explore alternatives. For instance, designers such as SuperFlux created several projects to explore the limits, risks, and potentials of our relationship with intelligent technologies. Similarly, Casnati et al. (Casnati et al., 2024) propose the notion of *provotypes* as a tangible or intangible design artifact whose main goal is to foster, encourage, and provoke discussions and reflections among different users and stakeholders. In the art world, Pinar Yoldas's work "*Ecologies of Excess"* (Yoldas, 2014) proposes a speculative biology project that imagines future life by creating new organisms which arise from the plastic in our oceans. Similarly, the artwork "I wanna deliver a dolphin" (Hasegawa, 2019) uses speculation to imagine a future humans can become a surrogate mother to endangered species, hence opening questions about ways of relation with the non-human. In academic research, an increasing number of scholars are turning to SF to reimagine alternative forms of inquiry. For instance, Freitas and Truman (2020) explore SF narratives as a lens to think about alternative research methods for studying posthuman ecologies. The special issue on "Imagining education

futures" (Hrastinski & Jandrić, 2023), instead, invites researchers in the field of education to write short SF narratives to imagine the future of education and open new paths for reflections. Similarly, Selwyn et al. (2020) use social science fiction to imagine how digital technologies might be used in a school of the future and what this might mean for the people whose lives are enmeshed with these technologies.

These diverse applications of SF underscore its potential to support critical reflection in three ways.

1) By fostering critical thinking: Through estrangement and unexpected perspectives, SF enables individuals to challenge assumptions and question prevailing norms.
2) By supporting critical understanding: Its socio-political dimensions help expose power structures, question the status quo, and disrupt dominant narratives (Iivari et al., 2022).
3) By encouraging critical imagining: SF opens creative spaces to envision alternative ways of living and relating to both human and non-human entities.

These possibilities are attracting an increasing number of educators and researchers to investigate and experiment with the possibilities of speculative fiction in education.

3.1. Speculative fiction as a tool to critically reflect on intelligent technologies with children

The use of SF in education is gaining traction among researchers as a tool for fostering critical reflection on a wide range of issues, including the climate crisis (Maxwell et al., 2019), LGBTQ + history (Millsap-Spears, 2018), bullying in schools (Ventä-Olkkonen et al., 2021, pp. 734–748), or the ethical complexities related to intelligent technologies (Burton et al., 2018; Hansen, 2021; Sharma et al., 2021).

This tendency is particularly evident in higher education. Burton et al. (2018) strongly advocate for the integration of science fiction in CS education to enhance reflexivity and provide richer perspectives on intelligent technologies. Specifically, they propose using science fiction narratives to examine the ethical dimensions of technology development, encouraging students to reflect on values, ethical dilemmas, and the broader consequences of design decisions. Similarly, Hansen (2021) proposed to use "optimistic science fiction" stories in STEM education as an instrument to offer different perspectives on technology and inspire students with beneficial examples of its ethical uses. Lukens and Disalvo (2011), on the other hand, emphasize the role of speculative design in fostering technological fluency by offering new ways to conceptualize technology and embedding critical thinking within the design process.

While these approaches have been explored extensively in higher education, fewer examples extend SF's educational potential to children's activities. Hardy (2019), for example, investigates how design fiction can support children's critical imagining by encouraging them to explore creative possibilities in technology design. She employs writing prompts based on design fiction scenarios, using them as scaffolding tools to help children push beyond the constraints of designing purely functional products. Similarly, Schaper and Ruiz (2023) introduced speculative questions about the future use of intelligent technologies after the COVID-19 lockdown and guided students in prototyping activities to design intelligent objects and agents that could be used in everyday situations in a near future in Spain. Finally, Sharma et al. (2021) note how speculative design can support children's learning about diversity and inclusion and help them advocate such important values in their future lives.

These perspectives highlight how SF can offer meaningful instruments to support *critical thinking, critical understanding,* and *critical imagining* when addressing complex phenomena. To better unpack the potential of SF for RAIL, in the following sections, we will use our reflections and empirical experiences to propose a framework oriented toward using SF to design appropriate resources to support children's critical reflection on intelligent technologies.

4. Using speculative fiction with children: account from three experiences

This article draws on self-examination and self-reflection on the authors' practices. Specifically, we will describe three cases in which the authors have explored different SF approaches to support children's critical reflections on intelligent technologies. The case studies were conducted independently as separate projects. We brought them together in this article to illustrate the breadth of educational contexts in which SF can be implemented, the range of educational goals it can achieve, and the variety of activities in which children could be engaged.

A total of eight researchers, with different backgrounds and at different stages of their academic careers, participated in these fieldworks. The wide range of employed techniques, combined with the diversity of researchers involved, created a rich foundation for stimulating debate and reflection on both the opportunities and limitations of these approaches. As well, these studies offered the unique chance for broader methodological considerations about SF for researching with children.

Our aim is not to provide a detailed account of the three cases (which are published elsewhere), but rather to use them as a foundation for reflecting on the methodological possibilities of SF in fostering reflective AI literacy. To bring coherence to the otherwise heterogeneous descriptions of the three cases, we will first provide a short description of each one. Subsequently, we will retrospectively examine our cases and discuss them in relation to the identified dimension of critical reflection.

4.1. Case 1: speculative fiction as an entry point to teaching teenagers about facial recognition technologies

Context and goals: In the first experience (Aki Tamashiroet al., 2021), we conducted an online workshop with two Danish female students (14–15 years) from a public primary school in (anonymized). One researcher facilitated the activities and the class teacher joined as a participant. The researcher documented observations in the form of field notes and conducted semi-structured interviews with the students at the end of each session (see Fig. 1)

The activities focused on exploring novel educational strategies in emerging technology education (van Mechelen et al., 2022), such as character design, role-playing, and speculative artifacts. Specifically, we investigated how a design-fiction approach could support students' reflections on their personal relationships with intelligent technologies in their everyday use.

Description of the activity: The activities were organized in three sessions with a duration of 2,5 h each. In the first session, the students were introduced to a design fiction storyline about their mission and explored applications of intelligent technologies. The design fiction scenario focused on a secret space mission of the "Intergalactic Confederation". The two students acted as representatives of Earth and had the task of observing and reporting potential risks and opportunities of new technologies that were introduced on other planets.

In this session, the researcher enabled the students to explore two different intelligent technologies. The first technology was an Augmented Reality tool that allows users to create face filters with doodles.[1] Within the fictional storyline, Scroobly was introduced as a tool that could change the appearance of any being, which allows blending in with other alien species. The second technology was a combination of the Teachable Machine[2]. (Carney et al., 2020) and Adacraft[3] (A Scratch-based coding environment that is compatible with ML extension blocks). By using these two technologies together, students

[1] http://www. scroobly.com.
[2] https://teachablemachine.withgoogle.com
[3] https://adacraft.org.

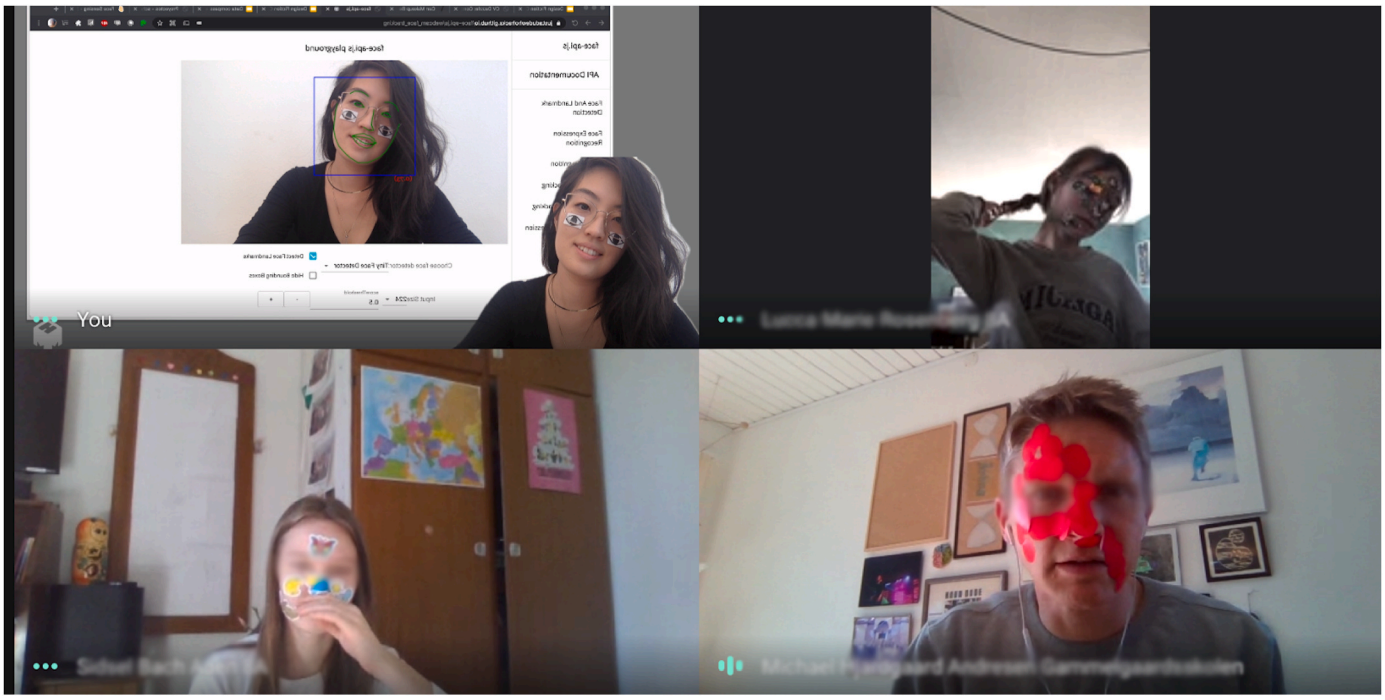


Fig. 1. Face Tricking activity engaged the students in understanding basic features of face-recognition systems.

were able to create an image classifier of objects that were important to them that they would fictionally bring to the space mission.

In the second session, we focused on ethical discussion and students' reflections on the accountability of stakeholders in the design and development of intelligent technologies. The students were introduced to Scratch-Lab's new face-sensing extension blocks.[4] They explored how to trick those facial recognition systems. We also facilitated a semi-structured discussion on different perspectives on the implications of this technology. Therefore, we employed the Data Compass (Bilstrup et al., 2022) activity. The goal was to introduce students to personal data as a concept that describes them as individuals. At the same time, it focuses on the value of this data for the ones who collect it. To do so, the students were presented with an editable version of the Data Compass which showed different example stakeholders in relation to facial recognition technologies and exemplary information of their private data. During the Data Compass activity, the students speculated about the use of data. For instance, they discussed how the triangulation of a person's age with other information can lead to knowing about the working years of this person in a specific institution (e.g. the government) and how this can give insights about the loyalty of the person towards his employer. Another example was when the students discussed how it would be possible to know someone's routine based on their patterns of listening to Spotify (it would be possible for example, to know if someone is at work, at the gym or at home depending on the playlist they were listening to). Afterward, students created their own characters along with their personality traits and perspectives on technology.

In the third session, we invited the students to speculate and design alternatives for intelligent technologies in the context of the design fiction story. Students were presented with inspiration from anti-surveillance artists, drag queens, and activists who experimented with tricking facial analysis for protests (e.g. Computer Vision Dazzle Camouflage[5]). Then, students were prompted to gather some materials, like stickers and makeup, to trick different facial analysis algorithms.[6] In the second part of the session, students used materials they had around the house to create speculative artifacts for the characters they created in session 2. One student made a jacket to switch between genders (made with a hoodie, as the jacket; some socks, as the pneumatic compartments to make some-one more masculine; and a corset, as a feature to make someone more feminine). The other student made a protection device for her character (made with an umbrella attached to a spray bottle, which conceptually would spray some substance to poison whoever was threatening the character). In the third part, students gave feedback and shared what they learned in all three sessions. In the third part, students gave feedback and shared what they learned in all three sessions.

4.1.1. *Children's reflections through the experience*

For the analysis, audio recordings of the semi-structured interviews were transcripted. Subsequently, transcripts and field notes were inductively analyzed to get a better understanding of how teenagers perceived the design-fiction approach in terms of (1) their engagement during the activities and (2) their understanding of societal implications of emerging technologies. The educational strategies showed to change students' perceptions about their exposure to intelligent technologies in their everyday lives, the stakeholders involved in the design of these technologies, and their potential risks in relation to data privacy and security. Particularly during the Data Compass activity, students speculated on how data, such as age and listening habits, could reveal personal information like workplace loyalty or daily routines. During the speculative artifacts activity, the students proposed using face-tricking strategies to hide as a criminal, for instance, using a smart jacket that can alter your appearance and gender or an intelligent umbrella that would defend its owner, in a "self-defense" attack.

4.2. *Case 2: speculative fiction to reflect on possible futures with social robots*

Context and goals: In the second experience (Rubegni et al. 2022),

[4] https://lab.scratch.mit.edu/face/.
[5] cvdazzle.com/.
[6] https://justadudewhohacks.github.io/face-api.js/webcam_face_tracking.

we run an activity with 60 children aged between 8 and 14 in Switzerland. Two researchers carried out the activity. The overall goal of the research was to investigate children's perceptions of social robots to better understand how children's hopes and fears influence their views of the future (anonymized).

Description of the activity: During the activity, the children were grouped into teams (up to 4) and were driven by the adult facilitator. As main scaffold resources, we used speculative scenarios that represent fictional situations in which children interact with a robot and/or an adult in everyday contexts. Specifically, the following scenarios were used: 1. Listening to problems, 2. Walking the dog, 3. Playing football, and 4. Co-reading at home. Each scenario was presented as a comic strip with three or four instances of the same activity (e.g., reading). For instance, in the "Co-reading at home" (Fig. 2) scenario there are four instances: in the first the child is reading to the robot, in the second the robot is reading to the child, in the third they read together, and in the fourth an adult reads to both. In the scenarios, we used the robot "Nao". Each scenario was the starting point to prompt children's reflections about their relationship with the technology represented.

At first, children had to give an individual rating of each instance of the scenario on a five-point Likert scale (1- I don't like it to 5 - I really like it). This was followed by a group discussion during which the facilitator used a set of prompts to stimulate children to talk about their thoughts, such as: *what do you think is happening here? What would you feel if you were that child? How would you change this scenario?* These prompts generated a discussion in which children shared their beliefs within the group. Each session lasted 60 min and we randomly created small teams of up to 4 children. We recorded the audio of each session that was transcribed and analyzed following a thematic (inductive and deductive) approach. Two people analyzed independently (in a blind mode) the transcription and then agreed on the coding. At first, we used an inductive approach and then a deductive one (Polit & Beck, 2014). In the data analysis, we denoted hope as statements around robots that appeared positive, optimistic, or beneficial to the children. In contrast, we coded fears as any statement that seemed anxious, worried, or unhappy about the use of social robots. Afterward, researchers agreed and refined the codes (49) that were subsequently collected in four themes that act as the four pillars of a conceptual model.

1. Boundaries and autonomy;
2. Cognitive features and comprehension skill;
3. Socioemotional features;
4. Materiality, physical and technical features.

While we use children's rating of the scenes to create some basic statistics of the children's preferences.

This exercise helped them to be open about their thoughts on the role of social robots and making meaning of these futuristic (but potential) realities. Data clearly showed how children showed ambivalence in the way they perceive robots in their lives. To represent this ambivalence, our model aims at using a dichotomous lens based on hopes and fears to make sense of children's perspectives of robots. The model shows the high level of complexity of children's thoughts about robots with the purpose of supporting the community to better understand how to challenge this topic.

Children's reflections through the experience: These proposed scenarios generated a rich discussion in which children shared their beliefs about social robots within the group. In particular, children expressed complex and often paradoxical fears and hopes related to the potential role of robots in their daily lives. Main fears addressed the perceived poor reliability of robots and their eventual misbehavior as well as the risks that robots may substitute people in caring for others. Instead, main hopes focused on the potential of having robots as companions when alone or having them as helpers in difficult or boring tasks.

The SF used in this study really helped in showing futuristic scenarios and in triggering the participants' thoughts by pushing them to feel in "the shoes" of the child in the strip. The children's strong reactions towards the scenarios followed by extensive discussions around specific topics of interest (e.g. the ability of the robot to fall in love) showed how much this method could be suitable for this type of research context (e.g. showing future realities) and questions.

4.3. Case 3: speculative fiction to reflect on communication technologies and their impact in children's lives

Context and goals: The third experience was run with 23 primary school children (10–12 years old) in a public school in Spain. Two researchers carried out the study. The activity focused on investigating children's perceptions, fears and hopes regarding mobile technologies and stimulating their reflections on the social impact of these devices. Our aim was to use this knowledge to extrapolate considerations and suggestions for guiding both media literacy initiatives as well as design practices.

Description of the activity: The activity was run in the school and was structured in 4 sessions of 90 min each. Two researchers facilitated all the sessions, together with the local teacher. A third researcher was accompanying the activity by taking field notes and audiovisual documentation. In the first session, we introduced the topic of the activity through a short introduction about the evolution of communication technologies into intelligent devices (e.g. smartphones, social media, etc.) during the last years and their role in mediating our relationships with others (especially during the COVID-19 lockdown). Subsequently, we divided the class into 4 groups and provided each one with a selection of provocative speculative illustrations by different artists that depicted different positive or negative possible scenarios regarding technology. We asked children to pick up one image for each of these statements: (1) an image with which you feel identified; (2) an image that worries you; (3) an image that represents how technology is changing our way of relating; (4) an image you like. Starting from the images, we involved children in a debate about how digital technologies are shaping our ways of relating to each other, identifying both negative and positive aspects. After the debate, we invited children to make two

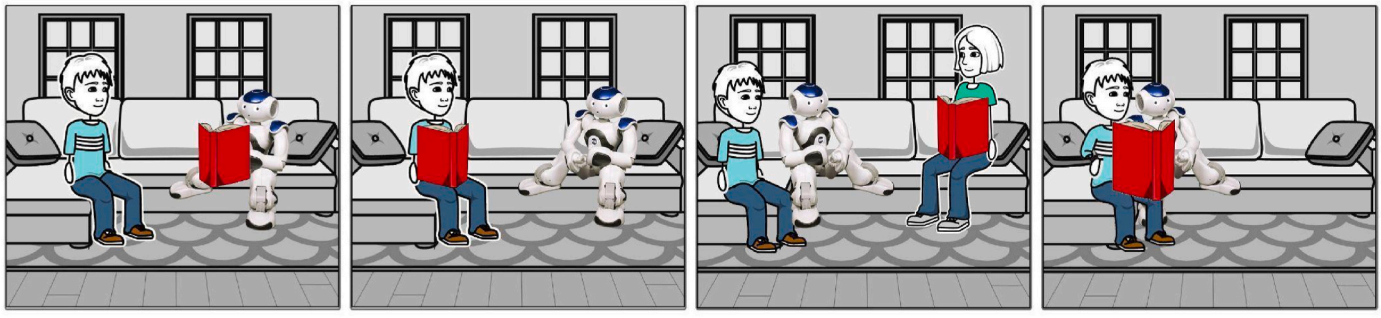


Fig. 2. An example of scenario strip: the co-reading.

drawings: (1) One depicting a future scenario where digital technologies evolve through aspects identified as positive; (2) One depicting a future scenario where digital technologies evolve through aspects identified as negative. Finally, children explained their drawings to others.

In the second session, we introduced the notion of speculative design. First, we did a recap of the previous session. Subsequently, we showed some references regarding speculative design artifacts produced by artists and designers. After that children, in groups of 4, received a paper bag containing a selection of craft materials. We invited all groups to use the available material to make a low-tech prototype of a technological artifact from the future that will be able to change our ways of relating to each other. At the end of the session, each group presented its ideas.

In the third and fourth sessions, children were invited to design and prototype their own speculative artifacts. Specifically, in the third session we presented the micro:bit toolkit and basic notions around its functioning. Children were therefore invited to augment the artifacts produced in the previous session through the use of micro:bit. In the last session, the children finished their prototypes and presented them to the rest of the class.

Children's reflections through the experience: The activity was analyzed by reviewing field notes and audiovisual materials. The main focus of the analysis addressed: (1) children's commentaries on mobiles technologies; (2) the design features of the produced prototypes. Since the first session, children's reflections were particularly rich and connected with their everyday experiences of digital technology mediation. Children, for instance debated around the experience of distance learning during the COVID-19 lockdown, the family conflicts related to the usage of digital technologies, the experience of feeling neglect because parents look at their phones and not at them, the social differences arising from the fact that some children own a mobile phone while other do not, the issue of addiction to digital technologies, etc. These conversations, elicited by the speculative illustrations we provided to them, were strongly related to children's fears and concerns regarding the effects that mobile technologies are having or may have in their lives. This tendency toward pointing out the risky aspects of technology influenced their design proposal. Most children, during the design activity, ended up designing artifacts that either describe dystopian futures (e.g. a device to radio-control teenager's movements because they are so addicted to the screen that they do not pay attention anymore to the physical space) or to control policies (i.e. artifacts whose role was to provide restrictions to technology usage such as a phone that can be used only for a limited amount of time otherwise you get physically punished).

5. Examining our practices through the lenses of critical reflection

In order to use our practices as "tools-to-think-with" we retrospectively examined them in relation to the identified dimensions of critical reflection: *critical thinking, critical understanding and critical imagining.*

5.1. Examining speculative fiction to support reflective AI literacy

In the three examples, SF served as a starting point to support children's reflections on intelligent technologies.

Specifically, in all cases, we employed SF to introduce the topic of the activity and to set the goals of children's reflection.

In the first case, SF was employed to engage students in critically reflecting on an already existing technology (e.g. facial recognition system) and to help them better understand its functionality, social implications, potential consequences, tensions, and risks arising from specific design choices. Therefore, the themes of reflection focused on data privacy, cybersecurity, and the accountability of technology stakeholders.

In the second case, we used SF to introduce a technology that does not yet exist in everyday life (e.g. social robots) but could potentially be implemented in the near future. This activity aimed to encourage children to speculate about and reflect on the socio-technical phenomena that could emerge with the implementation of this novel technology in everyday life. This was done by inviting them to focus on how social robots may change everyday interactions such as spending time with peers and family.

In the third case, rather than focusing on a specific technology, we used SF to explore children's relations with technologies that are well-established in their everyday lives (e.g. intelligent communication technologies). Children were invited to reflect on their personal experiences with well-known devices and on the social phenomena that may arise from their use. To this end, we asked them to bring to extreme consequences both positive and negative tendencies that they observe in their relation to these technologies.

These findings are consistent with literature on design futuring approaches (Kozubaev et al., 2020, pp. 1–15), on the potential of SF (Haraway, 2016), and of fabulation (Søndergaard et al., 2023, pp. 1693–1709). Across the three cases, SF showed to be suitable to engage children in reflecting about a broad spectrum of intelligent technologies, ranging from well-established devices to those not yet adopted. By allowing moving between different possible worlds, SF helped in freeding discussion from the limits of technological feasibility. Furthermore, in all cases, SF offered the possibility to inhabit the "what if" question, supporting world-making and speculative thinking. This feature allowed us to move smoothly between "what it is here now" and "what could be here in a near future", offering a rich scaffold for reflection. Moreover, by inherently encouraging the imagination of other worlds, SF expanded the scope of reflection beyond the technological artifact to encompass its social embeddedness (e.g. possible consequences of technology implementation in daily life). Finally, across all three cases, SF provided information in formats that can be more motivating for children than traditional ones, i.e. explaining how facial recognition works through a fictional narrative can be more engaging than just describing its technical features.

5.2. Examine our practices through the lens of critical thinking

In all three cases, the dimension of critical thinking was addressed. In the first case, for instance, self-examination of assumptions embedded in technology was conducted from a perspective similar to the work of Iversen et al. (2018) and Bilstrup et al. (2020, pp. 1211–1222), i.e. children were asked to assume the position of the designers and critically reflect on design decisions. Instead, in the second and third cases, children were invited to self-examine their own practices and beliefs either as current users or as potential users of the examined technologies.

To this end, SF-based resources showed to be useful especially for three main purposes. First, in all three cases, SF techniques provided children with provocative scenarios that stimulated debate, allowed them to confront their opinions and viewpoints with others. Second, the proposed scenarios allowed children to inhabit different roles in their reflections, helping them to adopt different perspectives. While other methodological approaches may also facilitate such role shifts (e.g., from user to designer), SF appears particularly helpful in facilitating the immersion needed to inhabit unusual roles (e.g. designers, policymakers). However, further research could deepen into understanding up to which point providing children with fictional contexts can really help them assume these roles (e.g. designers, policymakers, etc).

Third, in cases 2 and 3, the use of SF-based visual representations (i.e. the comic strips scenario in case 2 and the SF illustrations in case 3) functioned similarly to the technique of photo elicitation, i.e. using visual mediums to generate verbal discussion. In these cases, SF proved especially useful in providing provocative representations, which served as effective starting points for triggering dialogue and deepen reflection among children.

5.3. Examine our practices through the lens of critical understanding

In all three cases, the dimension of *critical understanding* was only marginally addressed. In case 1, the analysis of facial recognition systems and their relationship with non-normative gender representations provided an initial insight into the connection between technology design, prejudice, and discrimination and its role in perpetuating inequalities. However, the workshop did not fully explore the relation between these technological configurations and pre-existing systems of power and discrimination. Meanwhile, case 2 and case 3 focused mainly on how technologies could shape society, without explicitly aiming to an understanding of their socio-political dimensions and their connections with broader power structures.

Even if one project cannot cover all the possible dimensions of knowledge, we consider that for future research, it is necessary to place more emphasis on how SF can be used to support critical understanding. By reflecting on our practices through this lens, we can identify key shortcomings that may inform future studies. First, in all cases, the topics of reflection were framed from a predominantly techno-centric perspective, i.e. focusing on a specific device/technology instead of embracing the complexities of socio-technical ecosystems. Second, none of the cases explicitly integrated the goal of providing children with information about the socio-political or ecological impact of the addressed technologies. Third, none of the cases specifically embraced any political agenda aimed at supporting transformative practices.

5.4. Examine our practices through the lens of critical imagining

In the first and third cases, we attempted to use SF-based resources to foster critical imagining. However, our objectives were only partially achieved. Specifically, in both cases, SF-based artistic works were introduced to challenge children's habitual ways of thinking and subvert their usual ways of looking at emerging technologies. Our expectation was that these artistic examples would serve as inspiration, guiding the children's imaginative and creative processes during the design phase, where they were asked to conceptualize speculative prototypes. However, in both cases, we observed two predominant tendencies. First, children generally tended to imagine and focus on dystopian futures. Second, many of their designs were shaped by mainstream representations about technology use (e.g. control policies to avoid technology abuse). Both cases indicate that while offering inspirational resources can be valuable, is not enough to truly encourage the ability to challenge existing paradigms and the capacity to imagine unestablished and unconventional possibilities. Hence, even if SF has a long trajectory in offering compelling narratives to disrupt our ways of thinking, further research is needed regarding appropriate resources to support children's imagination in designing technologies that go beyond merely reproducing habitual ones.

5.5. Further considerations on the three cases

The examination of the three cases in relation to the dimensions of critical reflection (*critical thinking, critical understanding, and critical imagining*) show that, in all three cases, these dimensions are entangled within the practice and, in some circumstances, poorly exploited. This is certainly because we elaborated these dimensions "a posteriori" and we did not employ them in the design of the activities.

On the other hand, from a broader perspective, additional limitations can be found in the ways in which we framed children's participation in the three projects. In the three cases, as researchers, we decided on which themes we want children to reflect on. We consider that future research should more deeply explore children's interests and worries in order to define the topic under analysis.

At the same time, also in all three cases, SF served mainly as prompts that were presented to children. However, none of them deeply involved children in creating SF narratives or scenarios. Further research should therefore address how SF narrative creation could help reflective practices.

6. Opportunities and challenges in using speculative fiction to support children's critical reflection on intelligent technologies

Bringing together the revised literature, the reflection on our experiences and inspirations from SF narratives, we identified a set of opportunities and challenges related to using SF to address critical reflection in RAIL practices. In the next sections, we will present them in relation to the identified dimensions of critical reflection: critical thinking, critical understanding, and critical imagining. Their purpose is to guide practitioners and researchers in the tasks of designing and reflecting on RAIL activities for children.

6.1. Speculative fiction to foster critical thinking

Supporting critical thinking about intelligent technologies involve fostering children's capacity to question their pre-established assumptions, underlying premises, and dominant discourses around these technologies. Additionally, it entails encouraging children in self-reflection on their individual responsibilities as users and eventual future designers on intelligent technologies. As mentioned above, several RAIL projects address this dimension of critical reflection. Within this context, SF can offer specific opportunities to address these aspects in RAIL practices.

6.1.1. Opportunities

- SF prompts to provoke and question the commonplace: According to Dewey (1997), critical thinking occurs when a situation leads to "a state of perplexity, hesitation, doubt" that requires us to reconsider our ideas. In this regard, SF tradition is deeply rooted in the concept of estrangement to stimulate critical reflection. As shown in our cases, speculative prompts (e.g artworks, comic strips) helped children in questioning their own viewpoints and reconsider taken-for-granted ideas. In RAIL projects, both already existing and custom-designed SF narratives, scenarios and artistic works can serve as thought-provoking resources that, by generating unfamiliarity and offering children provoking ideas, can provide effective entry-points to trigger doubts and reflection.
- SF prompts to facilitate debate: Critical pedagogies have often employed dialogue and debate as meaningful resources to stimulate reflection (Freire, 1970; Lipman, 1976). Harri-Augstein (1991) suggests that self-reflective strategies alone are not enough, instead critical thinking depends on the negotiation of personal meanings through dialogue with others. In our case, the employed SF prompts (artworks, comic strips) showed to be useful resources to trigger the dialogue and stimulate a debate within groups of children around the defined topics. Also in this case, RAIL projects can employ either already existing and ad-hoc designed SF-based resources to kick-start dialogue about specific topics related to AI.
- SF prompts to inhabit different roles: Research on critical literacies highlights the importance of exposing children to diverse viewpoints to facilitate critical reflection (Pollard, 2019). In our cases, SF facilitated the immersion needed to inhabit other roles (e.g. to be a designer of the future) or other contexts. In this way, RAIL projects can use SF narratives to encourage children to adopt different roles and reflect from these standpoints, e.g. designers, future users, policymakers, etc. Furthermore, SF can help situate children in unfamiliar or thought-provoking contexts (e.g. living with a robot).
- SF prompts to move beyond technological feasibility: As suggested by Hardy (2019) and reinforced by our findings, SF allows for critical reflection on a broad spectrum of technologies, ranging from well-known devices to those that have not yet be implemented or even invented. This flexibility can be employed both to discuss

technologies as well as to design them, facilitating moving smoothly between "what is here now" and "what if could be here in a near future". For example, SF prompts can introduce children to a technology that does not exist yet or that is still not implemented in everyday life (e.g. a robot who will walk the dog). Similarly, we can use SF scenarios to help them in designing beyond the limits of what is doable.

6.1.2. Challenges

- Move beyond individual responsibility: RAIL projects that focus primarily on critical thinking often place a strong emphasis on individual ethical decision-making, both as technology users and as potential designers. This focus may run the risk of neglecting the complex socio-political landscape of intelligent technologies and overemphasize individual agency. Therefore, projects that integrate SF as a tool for critical reflection should be mindful of these limitations and strive to incorporate discussions about the broader socio-political dimensions of technological development.

6.2. Speculative fiction to foster critical understanding

Addressing critical reflection on AI from the perspective of critical understanding requires an explicit focus on the socio-political dimensions of these systems and a commitment to challenging and overturning social and political inequalities. However, research on RAIL with children that directly address this dimension remainsrelatively scarce. In this regard, we consider that SF can offer specific opportunities to more strongly support critical understanding in RAIL practices. Nonetheless, specific challenges also need to be addressed.

6.2.1. Opportunities

- SF to make visible the complexity of AI ecosystems: One important concern of AI literacy involves understanding AI's impact on society. To this end, there is a need in RAIL for approaches that allow expanding the scope of reflection beyond the technological artifacts to embrace its eco-social dimensions. As SF writer Le Guin (2005) highlights, imagining alternative worlds inherently implies imagining their values, technologies, structures of power, inhabitants, and the relationship between all these elements. This feature facilitates expanding the scope of reflection beyond the technological artifact to encompass and make visible the complex ecosystems that surrounds AI (e.g. the people that use them, those who produce and design them, the material resources needed, the influence of power and politics, etc.) (Crawford & Joler, 2018Crawford & Joler, 2018). In this regard, RAIL practices can take advantage of SF's world-building potential to introduce systemic knowledge and help children grasping these complex entanglements. This can be achieved through narratives, scenarios, and characters designed to illustrate this complexity and foster reflection on its interrelations.
- SF prompts to imagine the impact and consequences of technology development: As Burton et al. (2018) suggested, SF can be particularly useful for addressing the ethical dimensions of technology development and supporting reflection on the possible consequences of certain design decisions. In our cases, we saw that SF allowed children to see existing and future technologies through critical lenses and explore its possible developments and consequences. Within RAIL projects, SF can help in forecasting the impact and consequences of possible technology designs and developments. To this end, we consider that it could be very helpful to involve children more deeply in SF narrative creation and construction, since the task of having to imagine SF narratives related to AI will require them to deepen into the interplay between society and technologies.
- SF to support reflection of the role of power in shaping technology: As the AI Now Report (Kak & Myers West, 2023) emphasizes, there is a growing need to integrate the analysis of power distribution and resources allocation into reflection on AI. SF narratives can provide an engaging entry point for understanding how power shapes technological development—and, by extension, society. For example, Octavia E. Butler's novel *Parable of the Sower* (Butler, 2014) offers thoughtful insights on the interwoven relationships between power, ideology, technology development and social injustice. RAIL practices can take advantage of SF prompts about these relations as a starting point to scaffold discussion and imagine alternatives.
- SF to move beyond techno-centrism in RAIL: In all our cases, reflections on AI were largely framed through a techno-centric perspective, i.e. focusing on a specific device/technology rather than their broader societal entanglements. We suggest that SF's potential could be expanded by embracing more seriously the way in which SF literature allows re-thinking the relationships between humans and non-humans and dig into more-than-humans ecologies (Braidotti, 2013). Examples of them can be found in SF books such as *Binti* (Okorafor, 2018) where the writer imagines different couplings between different species and technologies, allowing thinking about wider ecologies and adopting other perspectives to technogenesis. This possibility, applied to the field of education, can eventually allow us to shift the topic of reflection from a specific technology to the broader systemic relations that technology-mediated ecologies are creating.
- SF to include marginalized viewpoints: As Parker (1996) highlights, critical understanding requires incorporating non-mainstream voices in the debate. SF can serve as a powerful tool in RAIL to amplify the voices of marginalized or underrepresented groups and to incorporate diverse perspectives. This could involve using SF to adopt alternative standpoints, such as exploring the perspectives of individuals who operate on the periphery of the AI ecosystem—e.g., those affected by AI biases, workers performing data annotation, or laborers in AI supply chains (e.g., mining workers extracting rare earth minerals).

6.2.2. Challenges

- Politicize RAIL: We consider that SF's potential can be further expanded by more explicitly acknowledging its political roots. Many SF cultural productions have been strongly characterized by political and vindictive intentions. A clear example of them is Afro-futurism, which explicitly denounces colonial history and addresses black people's liberation and creation of new stories and imaginaries about their past and future. Similarly, SF novels such as *Oryx and Crake* (Atwood, 2003)or *The Dispossessed* (Le Guin, 2015) are also well-known examples of political fiction. We argue that future research should critically examining the risks of maintaining a politically neutral stance, as failing to engage with these perspectives may diminish SF's potential to challenge dominant worldviews. In this sense, we consider that is an open challenge to reflect on political standpoints more broadly in RAIL research and educational practices.
- Support action: Critical understanding also entails the quest for involving students as active citizens, capable of taking action and participating in the transformation (Giroux, 1980). Addressing this goal is indeed challenging in educational practices that aim to foster critical reflection. In this context, we suggest that potential opportunities can be found in combining SF strategies with participatory approaches, to allow children and students to focus on topics that really matter to them.

6.3. Speculative fiction to foster critical imagining

Fostering critical imagining in RAIL involves encouraging children to break away from established norms, inhabit alternative ways of thinking and imagine new possibilities for understanding, designing, and using these technologies. Previous research on SF has already highlighted its

potential to offer alternative viewpoints and support the exploration of unconventional ideas. In this sense, the reflection of our practices contributes to this previous knowledge by further supporting it and identifying some additional opportunities and challenges.

6.3.1. Opportunities

- SF to provide strange viewpoints: As discussed earlier, SF provides useful resources to enable children assuming different roles and inhabit different perspectives. In our practices, SF-based artworks were particularly effective in presenting alternative viewpoints on specific technologies. This suggests that SF can be employed to encourage children to inhabit non-mainstream narratives or reflect from unconventional and "strange" standpoints in the technology ecosystem. For example, they might imagine the perspective of a non-human entity (a specific device, an algorithm, a lithium battery). An compelling example of this approach can be found in Ursula K. Le Guin's short text *Being taken for granite* (Le Guin, 2004) where, she explores the exercises of thinking and speaking from the position of different materials (e.g. granite, mud). Similarly, Kazuo Ishiguro's *Klara and the Sun* (Ishiguro, 2021) presents a dystopian narrative from the first-person perspective of a solar-powered AI robot, offering a controversial way to explore AI's societal role.
- SF to bring the "what if" to the extreme consequences: SF excels at leveraging the power of the "what if" question to reflect on tendencies and bring them to the extreme. In this regard, we suggest that RAIL projects can explore more deeply the power of the "what if" question by experimenting with the potential of branching narratives, where children can experiment with multiple possible futures. A well-known example is *Black Mirror: Bandersnatch*, which employs a **"choose your own adventure"** structure, allowing viewers to make decisions that shape the storyline. Such an approach could be adapted to help children explore the **ethical, social, and unintended consequences** of AI development through dynamic storytelling.
- SF to overcome technological solutionism: Morozov (2013) warns against the risks of technological solutionism, the tendency to frame complex social issues as neatly defined problems that can be solved solely through technological innovations. Often well-established design methods applied in education (e.g. design thinking) may run the risk of reinforcing this perspective by focusing on the idea of looking for a problem and trying to solve it through design. In this context, we suggest that SF-inspired design approaches can offer an alternative path: they encourage children to use design not just as a problem-solving tool, but to reimagine ways of living and coexisting on our planet. design methods leveraging SF can provide alternative paths to allow children to move from using design for solving problems to using design for imagining other possibilities of inhabiting our planet. To this end, SF can move beyond creating solutions toward exploring alternative futures, rethinking relationships with technology, and envisioning new ways of being in the world.
- SF to merge creativity with criticality: Supporting critical imagining about AI requires finding an appropriate balance between diverging for its own sake and diverging for imagining ethical and possible alternatives. In this sense, we consider that SF can help in putting together the dimensions of critical thinking, critical understanding and critical imagining by supporting creativity from a position that is critically informed.

6.3.2. Challenges

- Supporting children creating SF: SF is well recognized for its capacity to imagine alternative possibilities and to provide thought-provoking stories and images. However, relatively little research has focused on developing tools and resources to support children in imagining futures that extend beyond reproducing habitual or mainstream narratives. As observed in our cases, the tools we used were not particularly effective in freeing and opening children's imagination. This highlights the need for further research in designing resources that enable children to actively participate in creating SF. By doing so, SF creation can transcend its current association with experts—artists, designers, and writers—and instead become an accessible and guided process for children, empowering them to produce relevant, critical, and innovative SF narratives and artifacts.
- Move beyond dystopia: As our cases and previous research (Hansen, 2021) pointed out, young people's view of the future tends to be framed under the lenses of pessimism and catastrophe. Further research should explore the need for offering children resources that can enable more optimistic views about possible futures. Examples of them can be found in the work of Hansen (2021) or in narratives related to Solarpunk or Hope Punk movements (Reina-Rozo, 2021), where fiction is employed "as the springboard for guided ethical reflection can keep students focused on achieving ethicality rather than merely avoiding unethicality" (Hansen, 2021).

7. Conclusions

Our research aimed at understanding how SF-based resources can respond to the goals of supporting children's critical reflection on intelligent technologies and their impact on society. First, by revisiting relevant literature, we outlined three key dimensions of critical reflection (Critical Thinking, Critical Understanding, Critical Imagining) and pointed out the theoretical potential of SF in fostering critical engagement on intelligent technologies in children. Second, the examination of our own research studies allowed us to materialize these possibilities into concrete opportunities and challenges for educational activities. These are organized in three main dimensions we identified from the literature review: SF to foster critical thinking, SF to foster critical understanding, and SF to foster critical imagining.

Hence, this research provides two major contributions. On one hand, a theory-driven conceptual framework to address the definition of possible educational strategies to support key competences related to Reflective AI Literacy. On the other hand, it offers concrete methodological tools both for designing learning experiences for children as well as for opening novel research paths. Furthermore, additional elaboration of the identified opportunities can eventually serve to guide assessment and evaluation of children's competences in RAIL projects.

Selection and participation of children

Participants voluntarily participated in the different activities. Both parents and children agreed to participate by signing an informed consent. The authors do not have any potential conflict of interest.

All case studies shared key aspects regarding consent and the protection of participants' rights. In both instances, parents and legal guardians were fully informed about the objectives of the activities, and consent was obtained prior to the students' participation. In each study, the participants were made aware of the purpose of the research, and they were assured of their right to withdraw from the study at any point. Additionally, researchers in both studies acted as facilitators, ensuring that the participants felt comfortable and were not under any pressure. Data collected from the children were anonymized and securely stored, following strict privacy measures.

Nonetheless, there were some differences in the recruitment and organization processes. In the first study, students came from the same school and were selected by their class teacher based on their interest in emerging technologies. The school teacher played a key role in distributing the project information and consent forms to the students' legal guardians, and parents were invited to contact the research team for further information. In contrast, the second study was organized by the association Ated4Kids, which handled all aspects of the event's logistics and participant recruitment. The event was open to a broader group and

free of charge, with Ated4Kids responsible for informing parents and obtaining their consent, rather than the school or individual teachers. In the third case study, children belong to the same school and the whole class participated. The teacher distributed the consent form and information to parents prior to the study.

CRediT authorship contribution statement

Laura Malinverni: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Marie-Monique Schaper:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Elisa Rubegni:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Mariana Aki Tamashiro:** Writing – review & editing, Writing – original draft.

Ethical statement

Participants voluntarily participated in the different activities. Both parents and children agreed to participate by signing an informed consent. The studies were approved by the ethical committee within each institution in which each author was based at that moment. The authors do not have any potential conflict of interest.

Conflict of interest

The authors whose names are listed immediately above certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly and ChatGpt in order to proof-reading the final version of the manuscript and improve grammar issues. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Aki Tamashiro, M., Van Mechelen, M., Schaper, M. M., & Sejer Iversen, O. (2021). Introducing teenagers to machine learning through design fiction: An exploratory case study. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference* (pp. 471–475).

Atwood, M. (2003). *Oryx and Crake*. Bloomsbury.

Bilstrup, K. E. K., Kaspersen, M. H., & Petersen, M. G. (2020). Staging reflections on ethical dilemmas in machine learning: A card-based design workshop for high school students. *Dis 2020 - proceedings of the 2020 ACM designing interactive systems conference*. https://doi.org/10.1145/3357236.3395558

Bleecker, J. (2022). Design fiction: A short essay on design. *Science, Fact and Fiction. Machine Learning and the City: Applications in Architecture and Urban Design*, 561–578.

Braidotti, R. (2013). *The posthuman*. Wiley. https://www.wiley.com/en-es/The+Posthuman-p-9780745641584.

Bridle, J. (2022). Ways of being: Animals, plants. *Machines: The search for a planetary intelligence*. Penguin UK.

Brummelen, J., Heng, T., & Tabunshchyk, V. (2021). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum and development tools. In , *35, No. 17. In Proceedings of the AAAI Conference on Artificial Intelligence* (pp. 15655–15663).

Burton, E., Goldsmith, J., & Mattei, N. (2018). How to Teach Computer Ethics through Science Fiction: Science fiction in particular offers students a way to cultivate their capacity for moral imagination. *Communications of the ACM, 61*(8), 54–64. https://doi.org/10.1145/3154485

Butler, O. E. (2014). *Parable of the sower*. Hachette UK.

Carney, M., Webster, B., Alvarado, I., Phillips, K., Howell, N., Griffith, J., … Chen, A. (2020). Teachable machine: Approachable Web-based tool for exploring machine learning classification. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems* (pp. 1–8).

Casnati, F., Ianniello, A., & Romani, A. (2024). Provocation through narratives: New speculative design tools for human-non-human collaborations. In E. F. Zanella, G. Bosoni, E. Di Stefano, G. L. Iannilli, G. Matteucci, R. Messori, & R. Trocchianesi (Eds.), *Multidisciplinary aspects of design* (pp. 747–755). Nature Switzerland: Springer. https://doi.org/10.1007/978-3-031-49811-4_71.

Charisi, V., Malinverni, L., Schaper, M.-M., & Rubegni, E. (2020). Creating opportunities for children's critical reflections on AI, robotics and other intelligent technologies. *Proceedings of the 2020 ACM interaction design and children conference*. Extended Abstracts. https://doi.org/10.1145/3397617.3398063

Coenraad, M. (2022). "That's what techquity is": Youth perceptions of technological and algorithmic bias. *Information and Learning Sciences, 123*(7/8), 500–525. https://doi.org/10.1108/ILS-03-2022-0024

Crawford, K., & Joler, V. (2018). *Anatomy of an AI system. Anatomy of an AI system*.

Curry, A. (2018). *Worlds of Ursula K. Le Guin*.

de Andreotti, V. O. (2014). Soft versus critical global citizenship education. In E. S. McCloskey (Ed.), *Development education in policy and practice* (pp. 21–31). Palgrave Macmillan UK. https://doi.org/10.1057/9781137324665_2.

Dewey, J. (1997). How we think. *Courier Corporation*.

Dindler, C., Smith, R., & Iversen, O. S. (2020). Computational empowerment: Participatory design in education. *CoDesign, 16*(1), 66–80.

Druga, S., Vu, S. T., Likhith, E., & Qiu, T. (2019). Inclusive AI literacy for kids around the world. *Proceedings of FabLearn*. https://doi.org/10.1145/3311890.3311904, 2019.

Dunne, A., & Raby, F. (2013). *Speculative everything: Design, fiction, and social dreaming*. MIT Press.

Edelsky, C., & Cherland, M. R. (2006). A critical issue in critical literacy. In E. K. Cooper, & R. White (Eds.), *The practical critical educator* (pp. 17–33). Netherlands: Springer. https://doi.org/10.1007/1-4020-4473-9_2.

Freire, P. (1970). *Pedagogy of the oppressed*. New York: Herter and Herter.

Freitas, E., & Truman, S. (2020). New empiricisms in the anthropocene: Thinking with speculative fiction about science and social inquiry. *Qualitative Inquiry, 27*. https://doi.org/10.1177/1077800420943643

Fuchs, C. (2021). *Social media: A critical introduction* (pp. 1–440). Social Media.

Giroux, H. A. (1980). Critical theory and rationality in citizenship education. *Curriculum Inquiry, 10*(4), 329–366. https://doi.org/10.2307/1179823

Giroux, H. (2020). Critical pedagogy. In E. U. Bauer, U. H. Bittlingmayer, & A. Scherr (Eds.), *Handbuch Bildungs- und Erziehungssoziologie* (pp. 1–16). Springer Fachmedien. https://doi.org/10.1007/978-3-658-31395-1_19-1.

Hansen, K. S. (2021). Optimistic fiction as a tool for ethical reflection in STEM. *Journal of Academic Ethics, 19*(3), 425–439. https://doi.org/10.1007/s10805-021-09405-5

Haraway, D. J. (2016). *Staying with the trouble: Making kin in the chthulucene*. Duke University Press.

Hardy, A. (2019). Using design fiction to teach new and emerging technologies in england. *Technology and Engineering Teacher, 78*(4), 16–20.

Harri-Augstein, E. S. (1991). *Learning conversations: The self-organised learning way to personal and organisational growth/Sheila Harri-Augstein and Laurie F*. Thomas. London ; New York: Routledge. http://archive.org/details/learningconversa0000harr.

Hasegawa, A. (2019). I wanna deliver a dolphin…. https://aihasegawa.info/i-wanna-deliver-a-dolphin.

Herkert, J. R. (2005). Ways of thinking about and teaching ethical problem solving: Microethics and macroethics in engineering. *Science and Engineering Ethics, 11*(3), 373–385. https://doi.org/10.1007/s11948-005-0006-3

Hinrichsen, J., & Coombs, A. (2014). The five resources of critical digital literacy: A framework for curriculum integration. *Research in Learning Technology, 21*. https://doi.org/10.3402/rlt.v21.21334

Hitron, T., Orlev, Y., Wald, I., Shamir, A., Erel, H., & Zuckerman, O. (2019). Can children understand machine learning concepts?, 1-11 https://doi.org/10.1145/3290605.3300645.

Howell, N., F Schulte, B., Twigger Holroyd, A., Fatás Arana, R., Sharma, S., & Eden, G. (2021). Calling for a plurality of perspectives on design futuring: An un-manifesto. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–10.

Hrastinski, S., & Jandrić, P. (2023). Imagining education futures: Researchers as fiction authors. *Postdigital Science and Education, 5*(3), 509–515. https://doi.org/10.1007/s42438-023-00403-x

Iivari, N., Hartikainen, H., Ventä-Olkkonen, L., Sharma, S., Lehto, E., & Holappa, J. (2024). Exploring digital futures with children – implications for a future-oriented IS research. *ECIS 2024 Proceedings, 5*.

Iivari, N., Iversen, O. S., Smith, R. C., Schaper, M. M., Ventä-Olkkonen, L., Hartikainen, H., … Molin-Juustila, T. (2024). Transformative agency–the next step towards children's computational empowerment. *Proceedings of the 23rd annual ACM interaction design and children conference* (pp. 322–337).

Iivari, N., Sharma, S., Ventä-Olkkonen, L., Molin-Juustila, T., Kuutti, K., Holappa, J., & Kinnunen, E. (2022). Critical agenda driving child–computer interaction research—taking a stock of the past and envisioning the future. *International Journal of Child-Computer Interaction, 32*, Article 100408. https://doi.org/10.1016/j.ijcci.2021.100408

Ishiguro, K. (2021). *Klara and the sun*. Knopf.

Johnson, L., & Morris, P. (2010). Towards a framework for critical citizenship education. *Curriculum Journal, 21*(1), 77–96. https://doi.org/10.1080/09585170903560444

Kak, A., & Myers West, S. (2023). *AINow 2023 landscape: Confronting tech power*. AINow Institute. https://ainowinstitute.org/2023-landscape.

Kozubaev, S., Elsden, C., Howell, N., Søndergaard, M. L. J., Merrill, N., Schulte, B., & Wong, R. Y. (2020). Expanding modes of reflection in design futuring. *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–15). https://doi.org/10.1145/3313831.3376526

Le Guin, U. K. (2004). *The wave in the mind: Talks and essays on the writer, the reader, and the imagination*. Shambhala Publications, Inc.

Le Guin, U. K. (2005). A rant about «technology». https://www.ursulakleguin.com/a-rant-about-technology.

Le Guin, U. K. (2015). *The dispossessed*. Hachette UK.

Lin, P., Van Brummelen, J., Lukin, G., Williams, R., & Breazeal, C. (2020). Zhorai: Designing a conversational agent for children to explore machine learning concepts. *Proceedings of the AAAI Conference on Artificial Intelligence, 34*(9), 13381–13388. https://doi.org/10.1609/aaai.v34i09.7061

Lipman, M. (1976). Philosophy for children. *Metaphilosophy, 7*(1), 17–39.

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of the 2020 CHI conference on human factors in computing systems. CHI '20: Proceedings of the 2020 CHI conference on human factors in computing systems*. https://dl.acm.org/doi/10.1145/3313831.3376727.

Luke, A. (2000). Critical literacy in Australia: A matter of context and standpoint. *Journal of Adolescent & Adult Literacy, 43*(5), 448–461.

Lukens, J., & Disalvo, C. (2011). Speculative design and technological fluency. *International Journal of Learning and Media, 3*, 23–40. https://doi.org/10.1162/IJLM_a_00080

Maxwell, D., Pillatt, T., Edwards, L., & Newman, R. (2019). Applying design fiction in primary schools to explore environmental challenges. *The Design Journal, 22*(sup1), 1481–1497. https://doi.org/10.1080/14606925.2019.1594972

McLaughlin, T. H. (1992). Citizenship, diversity and education: A philosophical perspective. *Journal of Moral Education, 21*(3), 235–250. https://doi.org/10.1080/0305724920210307

Mezirow, J. (1987). *Fostering critical reflection in adulthood: A guide to transformative and emancipatory learning* (pp. 214–216). Interpreting.

Millsap-Spears, C. (2018). "Teaching with trek": Star trek, the LGBTQ+ community, and college composition. In E. S. Rabitsch, M. Gabriel, W. Elmenreich, & J. N. A. Brown (Eds.), *Set phasers to teach!: Star trek in research and teaching* (pp. 95–105). Springer International Publishing. https://doi.org/10.1007/978-3-319-73776-8_9.

Morozov, E. (2013). *To save everything, click here: The folly of technological solutionism*. PublicAffairs.

Murray, J. B., & Ozanne, J. L. (2006). Rethinking the critical imagination. In *Handbook of qualitative research methods in marketing*. Edward Elgar Publishing.

Okorafor, N. (2018). *Binti: The night masquerade*. Macmillan.

Oziewicz, M. (2017). *Speculative fiction. En oxford research encyclopedia of literature*.

Pandya, J., & Avila, J. (Eds.). (2013). *Moving critical literacies forward: A new look at praxis across contexts*. Routledge. https://doi.org/10.4324/9780203521861.

Pangrazio, L. (2016). Reconceptualising critical digital literacy. *Discourse: Studies in the Cultural Politics of Education, 37*(2), 163–174. https://doi.org/10.1080/01596306.2014.942836

Parisier, E. (2012). *The filter bubble: How the new personalized web is changing what we read and how we think*. New York, NYUnited States: Penguin Books375 Hudson Street.

Parker, W. (1996). "Advanced" ideas about democracy: Toward a pluralist conception of citizen education. *Teachers College Record, 98*(1), 104–125.

Payne, B. H. (2020). *Can my algorithm be my opinion?: An AI+ ethics curriculum for middle school students [PhD Thesis]*. Doctoral dissertation, Massachusetts Institute of Technology.

Pollard, B. (2019). Utilizing a critical literacy framework to discuss issues of power and privilege with elementary students. *Intersections: Critical Issues in Education, 3*(2). https://digitalrepository.unm.edu/intersections/vol3/iss2/5.

Reina-Rozo, J. D. (2021). Art, energy and technology: The Solarpunk movement. *International Journal of Engineering, Social Justice, and Peace, 8*(1). https://doi.org/10.24908/ijesjp.v8i1.14292. Article 1.

Rubegni, E., Malinverni, L., & Yip, J. (2022). "Don't let the robots walk our dogs, but it's ok for them to do our homework": children's perceptions, fears, and hopes in social robots. In *Proceedings of the 21st Annual ACM Interaction Design and Children Conference* (pp. 352–361).

Ruppert, J., Velazquez-Ramos, D., Roque, R., & Shapiro, B. R. (2023). Taking play and tinkering seriously in AI education: Cases from drag vs AI teen workshops. *Learning, Media and Technology, January*, 1–15. https://doi.org/10.1080/17439884.2022.2164300

Schaper, M.-M., Malinverni, L., & valero, c (2020). Robot presidents: Who should rule the world? Teaching critical thinking in AI through reflections upon food traditions. *Proceedings of the 11th nordic conference on human-computer interaction: Shaping experiences* (pp. 1–4). Shaping Society. https://doi.org/10.1145/3419249.3420085

Schaper, M.-M., & Ruiz, A. (2023). Three recommendations to engage at-risk students in critical reflection on intelligent technologies through remote learning. *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. https://doi.org/10.1145/3544549.3573862

Schaper, M.-M., Smith, R. C., Tamashiro, M. A., Van Mechelen, M., Lunding, M. S., Bilstrup, K.-E. K., Kaspersen, M. H., Jensen, K. L., Petersen, M. G., & Iversen, O. S. (2022). Computational empowerment in practice: Scaffolding teenagers' learning about emerging technologies and their ethical and societal impact. *International Journal of Child-Computer Interaction, 34*, Article 100537. https://doi.org/10.1016/j.ijcci.2022.100537

Selwyn, N., Pangrazio, L., Nemorin, S., & Perrotta, C. (2020). What might the school of 2030 be like?: An exercise in social science fiction. *Learning, Media and Technology, 45*(1), 90–106. https://doi.org/10.1080/17439884.2020.1694944

Sharma, S., Hartikainen, H., Ventä-Olkkonen, L., Iivari, N., Eden, G., Kinnunen, E., Holappa, J., Kinnula, M., Molin-Juustila, T., Okkonen, J., Kotilainen, S., Iversen, O., & Fatas, R. (2021). In *Pursuit of inclusive and diverse digital futures: Exploring the potential of design fiction in education of children* (pp. 219–248). Interaction Design and Architecture(s). https://doi.org/10.55612/s-5002-051-010.

Sharma, S., Kinnula, M., Iivari, N., Ventä-Olkkonen, L., Hartikainen, H., Durall, E., Molin-Juustila, T., Okkonen, J., Kotilainen, S., Sawhney, N., Eden, G., & Monga, C. (2022). Age against the machine: A call for designing ethical AI for and with children. *Adjunct proceedings of the 2022 nordic human-computer interaction conference* (pp. 1–6). https://doi.org/10.1145/3547522.3547702

Søndergaard, M. L. J., Campo Woytuk, N., Howell, N., Tsaknaki, V., Helms, K., Jenkins, T., & Sanches, P. (2023). Fabulation as an approach for design futuring. *Proceedings of the 2023 ACM designing interactive systems conference* (pp. 1693–1709).

Touretzky, D., Gardner-McCune, C., Martin, F., & Seehorn, D. (2019). Envisioning AI for K-12: What should every child know about AI? *Proceedings of the AAAI Conference on Artificial Intelligence, 33*(1). https://doi.org/10.1609/aaai.v33i01.33019795. Article 01.

Vakil, S. (2018). Equity in computer science education. *Harvard Educational Review, 88*(1), 26–53.

Ventä-Olkkonen, L., Iivari, N., Sharma, S., Molin-Juustila, T., Kuutti, K., Juustila-Cevirel, N., Kinnunen, E., & Holappa, J. (2021). *Nowhere to now-here: Empowering children to reimagine bully prevention at schools using critical design fiction: Exploring the potential of participatory* (pp. 734–748). Empowering Design Fiction in Collaboration with Children. https://doi.org/10.1145/3461778.3462044

Westheimer, J., & Kahne, J. (2004). What kind of citizen? The politics of educating for democracy—joel westheimer, joseph Kahne, 2004. *American Educational Research Journal, 41*(2), 237–269.

Wilde, D., Vallgårda, A., & Tomico, O. (2017). Embodied design ideation methods. *Proceedings of the 2017 CHI conference on human factors in computing systems*. https://dl.acm.org/doi/abs/10.1145/3025453.3025873.

Winner, L. (1980). Do artifacts have politics? *Dædalus, 109*(1), 121–136.

Yoldas, P. (2014). *Ecologies of excess*.